RAG Learning

Edexcel A-Level Chemistry — Core Practicals

Table of Contents

1. CP1: Molar Mass of a Volatile Liquid or Gas
2. CP2: Acid-Base Titration
3. CP3: Enthalpy of Neutralisation (Calorimetry)
4. CP4: Enthalpy of Combustion of an Alcohol
5. CP5: Rate of Reaction — Initial Rates (Iodine Clock)
6. CP6: Rate of Reaction — Continuous Monitoring (Gas Collection)
7. CP7: Qualitative Analysis (Flame Tests, Cation/Anion Tests)
8. CP8: Electrolysis of Aqueous Solutions
9. CP9: K_{sp} of Calcium Hydroxide by Titration
10. CP10: Electrochemical Cells and EMF
11. CP11: Temperature and Rate (Thiosulfate + HCl)
12. CP12: Factors Affecting K_c
13. CP13: Prepare a Pure Organic Solid and Test Purity
14. CP14: Prepare a Pure Organic Liquid (Distillation/Reflux)
15. CP15: Tests to Identify Functional Groups
16. CP16: pH Changes During Acid-Base Titrations

CP1: Determine the Molar Mass of a Volatile Liquid or Gas

Background Theory

The molar mass (M_r) of a volatile liquid can be determined experimentally using the **ideal gas equation: pV = nRT**. A known mass of the volatile liquid is vaporised at a known temperature and pressure, and the volume of the resulting gas is measured. Since n = mass/M_r, the equation can be rearranged to M_r = mRT/(pV). Two methods are commonly used: the **gas syringe method** (inject liquid into a heated gas syringe) and the **Dumas method** (heat liquid in a flask with a pinhole until it all vaporises, then weigh the condensed vapour).

$$pV = nRT \quad \rightarrow \quad n = pV/RT \quad \rightarrow \quad M_r = mRT/(pV)$$

Where: p = pressure (Pa), V = volume (m^3), n = moles, R = 8.314 J mol^{-1} K^{-1}, T = temperature (K), m = mass (g)

Method (Gas Syringe)

1. Place a gas syringe (with a self-sealing rubber septum at one end) inside an oven set to a known temperature well above the boiling point of the liquid. Allow to reach thermal equilibrium. Record the temperature (T) in Kelvin and atmospheric pressure (p) in Pa.
2. Weigh the hypodermic syringe containing a small volume (~0.2 cm^3) of the volatile liquid. Record mass to ±0.001 g.
3. Inject the liquid through the septum into the hot gas syringe. The liquid vaporises immediately. Re-weigh the hypodermic syringe.
4. Mass of liquid injected = initial mass − final mass.
5. Once the gas syringe has reached thermal equilibrium, record the volume of gas (V) in cm^3. Convert to m^3 (1 cm^3 = 1 x 10^{-6} m^3).
6. Calculate: n = pV/(RT). Then M_r = mass/n.

Method (Dumas)

1. Place a few drops of the volatile liquid in a round-bottomed flask. Cover the opening with aluminium foil pierced with a **single pin-hole**. The pinhole allows vapour to escape but prevents air from entering rapidly.
2. Place the flask in a **boiling water bath** (100 degrees C / 373 K). As the liquid vaporises, excess vapour escapes through the pinhole until only vapour at atmospheric pressure remains in the flask.

3. When no more vapour is seen escaping, remove the flask and allow it to cool. The vapour condenses back to liquid.
4. Weigh the flask + condensed liquid + foil. Then weigh the empty, dry, clean flask + foil. Mass of vapour = difference.
5. Measure the volume of the flask by filling with water and measuring the volume of water. This gives V.
6. T = temperature of the water bath (in K); p = atmospheric pressure (in Pa).
7. Calculate $M_r = mRT/(pV)$.

Safety

- Volatile liquids are **highly flammable** — no naked flames; use electric oven/water bath.
- Handle hypodermic syringes with care (sharp needle — risk of needlestick injury).
- Oven is hot — use heat-resistant gloves.
- Work in a well-ventilated area (volatile vapours can be irritating).

Unit Conversions (Essential)

Quantity	Given In	Convert To	Conversion
Volume	cm^3	m^3	Multiply by 10^{-6}
Temperature	degrees C	K	Add 273
Pressure	atm or kPa	Pa	1 atm = 101,325 Pa; 1 kPa = 1000 Pa

Error Analysis

Source of Error	Type	Improvement
Gas syringe friction (reads low volume)	Systematic (M_r too high)	Tap syringe gently to ensure free movement; use well-lubricated syringe
Temperature not uniform in oven	Systematic	Allow plenty of time for thermal equilibrium; place thermometer near the syringe

Liquid not fully vaporised	Systematic (volume too low, M_r too high)	Ensure oven temperature is well above the boiling point of the liquid
Vapour escaping around septum	Systematic (volume too low)	Check the septum seal is airtight before injecting
Ideal gas assumption	Systematic	Real gases deviate from ideality, especially at high pressures and low temperatures. The assumption is more valid at high T and low P. Acknowledge this limitation
Dumas: vapour escaping after removal from water bath	Systematic (mass too low)	Quickly seal the pinhole as soon as the flask is removed; cool rapidly

Exam Tip: Always convert **all units** before substituting into pV = nRT: volume to m^3, temperature to K, pressure to Pa. The ideal gas equation assumes no intermolecular forces and that the gas molecules have negligible volume — these assumptions break down at high pressures and low temperatures. Experimental M_r values are typically slightly higher than true values because (1) some vapour may condense before the volume is read, and (2) gas syringe friction causes the volume to read low.

CP2: Acid-Base Titration (Finding Concentration)

Background Theory

See the AQA RP1 entry for comprehensive titration theory, method, indicator selection, and error analysis. The Edexcel specification uses the same technique. Key points specific to Edexcel: the practical endorsement requires you to demonstrate competence in pipetting, burette use, and concordant titre determination. You should be able to calculate concentrations, moles, and masses from titration data using n = cV and the balanced equation.

Key Practical Points

- Burette readings to **2 decimal places** (nearest 0.05 cm^3)
- Concordant results within **0.10 cm^3**
- **Rinse** the burette with the acid it will contain; **rinse** the pipette with the alkali. Do **NOT** rinse the conical flask with alkali (a wet conical flask does not affect the result because it does not change the number of moles of alkali)
- Use a **white tile** to see colour changes clearly
- Add drop-wise (or half-drops) near the end-point for accuracy
- Indicator choice: methyl orange for strong acid/strong base or strong acid/weak base; phenolphthalein for weak acid/strong base

> **Exam Tip:** A common question asks why a wet conical flask does NOT cause an error: the extra water does not change the moles of alkali already in the flask, so the same volume of acid is still needed. But a wet pipette or wet burette WOULD cause an error because the water dilutes the solution, changing its effective concentration.

CP3: Enthalpy of Neutralisation (Calorimetry)

Background Theory

See the AQA RP2 entry for comprehensive calorimetry theory, graphical extrapolation method, assumptions, and error analysis. Key additional Edexcel points:

- The standard enthalpy of neutralisation for strong acid + strong base (e.g. HCl + NaOH) is approximately **−57.1 kJ mol^{-1}** because the net ionic equation is the same: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
- For reactions involving weak acids or bases, the measured ΔH is less exothermic because some energy is used to fully dissociate the weak acid/base (bond breaking is endothermic)
- Always use the **graphical extrapolation method** to find the corrected ΔT: plot temperature vs time, extrapolate the pre-mixing and post-mixing lines back to the time of mixing

$$q = mc\Delta T \quad \Delta H = -q/n \quad (\text{kJ mol}^{-1})$$

Exam Tip: Experimental values are always **less exothermic** than literature values due to heat loss. The two key assumptions: c = 4.18 J g^{-1} K^{-1} and density = 1.0 g cm^{-3}. For neutralisation of a weak acid, ΔH is less than 57.1 kJ mol^{-1} because energy is needed to fully ionise the weak acid first.

CP4: Determine the Enthalpy of Combustion of an Alcohol

Background Theory

The **enthalpy of combustion ($\Delta_c H$)** is the enthalpy change when one mole of a substance is completely burned in excess oxygen under standard conditions. This practical uses a simple **spirit burner calorimeter**: the alcohol burns and heats a known mass of water above it. By measuring the temperature rise of the water and the mass of alcohol burned, $\Delta_c H$ can be calculated. Experimental values are always significantly less exothermic than data book values, primarily due to substantial heat loss to the surroundings. Across a homologous series of alcohols, $\Delta_c H$ increases approximately linearly with each additional CH_2 group (~650 kJ mol^{-1} per CH_2), because each additional CH_2 provides additional C–H and C–C bonds to be broken and additional C=O and O–H bonds to be formed.

Apparatus

- Spirit burner containing the alcohol to be tested (e.g. methanol, ethanol, propan-1-ol, butan-1-ol)
- Copper calorimeter (or metal can — copper has high thermal conductivity, transferring heat efficiently to water)
- Thermometer (±0.5 degrees C) or temperature probe
- Measuring cylinder (100 cm^3), balance (±0.01 g)
- Draught shield (to reduce heat loss from wind), clamp stand, stirring rod

Method

1. Measure **100 cm^3** of cold water into the copper calorimeter. Record the **initial temperature** (T_1).
2. Weigh the spirit burner (with cap on) and record the mass to ±0.01 g (m_1).
3. Position the calorimeter above the spirit burner. Surround with the draught shield.
4. Light the spirit burner. Stir the water gently with the stirring rod throughout.
5. When the temperature has risen by approximately **20 degrees C**, extinguish the flame by replacing the cap. Record the **maximum temperature** (T_2).
6. Immediately re-weigh the spirit burner (m_2).
7. Calculate:
   - $\Delta T = T_2 - T_1$
   - Mass of fuel burned = $m_1 - m_2$
   - $q = mc\Delta T$ (m = 100 g water, c = 4.18 J g^{-1} K^{-1})
   - Moles of fuel = $(m_1 - m_2) / M_r$
   - $\Delta_c H = -q / n$ (kJ mol^{-1})
8. Repeat for different alcohols in the same homologous series.

Safety

- Alcohols are **highly flammable** — keep containers away from the flame; cap the burner promptly.
- Spirit burner flames can be **almost invisible** (especially methanol) — handle carefully; assume the burner is lit until confirmed otherwise.
- Hot apparatus — use tongs and a heatproof mat.

Analysis

- Plot **$\Delta_c H$ (y-axis)** against **number of carbon atoms (x-axis)**. Expect a linear trend: each additional CH_2 group adds ~650 kJ mol^{-1} to the combustion enthalpy.

- Compare experimental values with data book values. Experimental values will be **much less exothermic** (typically 50–70% of literature values) due to the dominant source of error: heat loss.

Error Analysis

Source of Error	Type	Effect on $\Delta_c H$	Improvement
Heat loss to surroundings (dominant error)	Systematic	ΔT too small; $\Delta_c H$ less exothermic	Better insulation; draught shield; use a bomb calorimeter (industrial standard)
Incomplete combustion (soot/CO produced)	Systematic	Less heat released per mole; $\Delta_c H$ less exothermic	Ensure adequate air supply; trim and clean the wick; observe flame colour (yellow/sooty = incomplete)
Evaporation of alcohol from hot wick after extinguishing	Systematic	Mass loss overstated; n too high; $\Delta_c H$ appears less exothermic	Cap the burner promptly; re-weigh quickly
Heat absorbed by calorimeter, stirrer, and thermometer	Systematic	ΔT of water understated	Use a copper calorimeter (low heat capacity); determine a calorimeter constant
Water evaporation during heating	Systematic	Mass of water decreases; less water to heat	Use a lid on the calorimeter to reduce evaporation

Exam Tip: Experimental $\Delta_c H$ values are always **less exothermic** (smaller magnitude) than data book values. The main reason is **heat loss to the surroundings**. Incomplete combustion also contributes (producing CO and C instead of only CO_2, releasing less energy). The **linear trend** across the homologous series shows that each additional CH_2 group contributes approximately the same amount of energy. Use bond enthalpy calculations to explain the

theoretical trend: breaking C–H and C–C bonds (endothermic) and forming C=O and O–H bonds (exothermic).

CP5: Rate of Reaction — Initial Rates Method (Iodine Clock)

See the AQA RP8 entry for detailed theory, method, order determination, and error analysis. The Edexcel specification uses the same iodine clock technique. Key points: Rate = 1/t; keep total volume constant; plot rate vs [X] to determine order (straight line through origin = first order; curve = second order, confirmed by plotting rate vs $[X]^2$).

Exam Tip: To determine order from experimental data: if doubling [X] doubles the rate, the reaction is **first order** with respect to X. If doubling [X] quadruples the rate, it is **second order**. If changing [X] has no effect on rate, it is **zero order**. The overall rate equation is Rate = $k[A]^m[B]^n$, where m + n = overall order. The rate constant k has units that depend on the overall order.

CP6: Rate of Reaction — Continuous Monitoring (Gas Collection)

See the AQA RP7 entry for detailed theory, gas syringe/mass loss methods, and analysis of volume-time graphs. Key Edexcel points: the initial rate = gradient of the tangent at t = 0. Compare curves for different concentrations, surface areas, or temperatures. The curve levels off when one reactant is consumed.

Exam Tip: On a volume-time graph: steeper initial gradient = faster rate. If the **same moles of limiting reagent** are used, all curves reach the **same final volume**. Increasing

concentration makes the reaction faster (steeper curve) but does not change the total amount of product if the limiting reagent amount is unchanged.

CP7: Qualitative Analysis (Flame Tests, Cation and Anion Tests)

See the AQA RP4 entry for comprehensive cation and anion identification tables, flame test colours, precipitation reactions with NaOH and NH_3, and the key distinguishing tests (Al^{3+} vs Mg^{2+} with excess NaOH; Cu^{2+} with excess NH_3).

Exam Tip: Edexcel frequently asks you to identify an unknown salt from a set of test results. Approach systematically: (1) flame test narrows down the cation, (2) NaOH/NH_3 precipitation confirms it, (3) anion tests ($AgNO_3$ for halides, $BaCl_2$ for sulfate, HCl for carbonate) identify the anion. Always acidify with HNO_3 before adding $AgNO_3$ or $BaCl_2$ to remove interfering ions.

CP8: Electrolysis of Aqueous Solutions

Background Theory

Electrolysis is the decomposition of an ionic compound (electrolyte) using an electric current. In aqueous solutions, there are always **competing ions** at each electrode: the ions from the dissolved salt compete with H^+ and OH^- from the dissociation of water. The ion that is **preferentially discharged** depends on its position in the electrochemical series (electrode potential) and its concentration.

Discharge Rules for Aqueous Solutions

Electrode	Process	Rule
Cathode (−)	Reduction of cations	The **least reactive** cation is discharged first (most positive E^o). Metal ions less reactive than hydrogen (e.g. Cu^{2+}, Ag^+) are deposited as metal. More reactive metal ions (e.g. Na^+, K^+, Ca^{2+}) remain in solution and H^+ is discharged instead (producing H_2 gas)
Anode (+)	Oxidation of anions	**Halide ions** (Cl^-, Br^-, I^-) are discharged in preference to OH^- if present in reasonable concentration (producing Cl_2, Br_2, or I_2). If no halide is present, OH^- is discharged, producing O_2 gas. Exception: very dilute halide solutions may discharge OH^- instead

Apparatus

- Beaker (250 cm^3), two carbon (graphite) or inert platinum electrodes (or copper electrodes for $CuSO_4$ experiments)
- DC power supply (6 V) or battery, connecting wires with crocodile clips
- Ammeter (optional, to monitor current), rheostat (optional, to control current)
- Various electrolyte solutions: $CuSO_4$(aq), NaCl(aq), dilute H_2SO_4(aq)
- Inverted test tubes (for collecting gases over electrodes), splints, damp litmus paper

Method

1. Pour the electrolyte solution into the beaker to a depth that covers both electrodes.
2. Place two graphite electrodes in the solution, held apart by clamps or an electrode holder.
3. Connect the electrodes to the DC power supply. Ensure you know which is the anode (+) and which is the cathode (−).
4. Switch on and observe both electrodes for 5–10 minutes. Record observations (colour changes, bubbles, deposits, smells).
5. Test any gases produced: **O_2** relights a glowing splint; **H_2** gives a squeaky pop with a burning splint; **Cl_2** bleaches damp litmus paper (turns it white).
6. Repeat with different electrolytes. Record and compare products.

Expected Results

Electrolyte	Cathode Product	Anode Product	Explanation
$CuSO_4$(aq) with graphite electrodes	Copper (brown/pink solid deposits)	O_2 gas (bubbles)	Cu^{2+} less reactive than H^+, so Cu deposited. SO_4^{2-} too stable; OH^- discharged instead: $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
$CuSO_4$(aq) with **copper** electrodes	Copper deposits	Copper dissolves	Anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$. Cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$. Net: copper transfers from anode to cathode. This is **copper purification/electroplating**
Concentrated NaCl(aq)	H_2 gas (squeaky pop)	Cl_2 gas (bleaches litmus)	Na^+ too reactive; H^+ discharged. Cl^- discharged in preference to OH^- at high concentration
Dilute H_2SO_4(aq)	H_2 gas	O_2 gas	H^+ reduced at cathode. OH^- oxidised at anode (no halide present). Net: electrolysis of water

Safety

- **Chlorine gas is toxic** — work in a fume cupboard when electrolyzing NaCl. Small quantities only.
- Do not touch electrodes while the power supply is on (electric shock risk).
- $CuSO_4$ is harmful — avoid skin contact; wear gloves.
- H_2 is flammable — keep away from naked flames.

Error Analysis

Source of Error	Type	Improvement

Gas dissolving in solution before collection	Systematic	Collect gas in inverted test tubes filled with solution and placed directly over the electrode
Products at anode and cathode mixing (especially Cl_2 and H_2)	Systematic	Use a diaphragm or porous pot to separate the two electrode compartments (industrial chlor-alkali process uses a membrane cell)
Electrode contamination	Systematic	Clean electrodes before use; use new graphite rods or sand copper electrodes

Exam Tip: At the cathode: the **least reactive** cation is discharged (most positive E^{o}). At the anode: **halide ions are preferentially discharged** over OH^- (if present in sufficient concentration). With **copper electrodes** in $CuSO_4$: the copper anode dissolves and copper deposits at the cathode — the solution concentration stays constant. This is the basis of **copper purification** (impure copper anode, pure copper cathode, $CuSO_4$ electrolyte).

CP9: Determine K_{sp} of Calcium Hydroxide by Titration

Background Theory

The **solubility product (K_{sp})** of a sparingly soluble salt is the product of the concentrations of its ions in a saturated solution, each raised to the power of their stoichiometric coefficients. For calcium hydroxide:

$$Ca(OH)_2(s) \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq) \qquad K_{sp} = [Ca^{2+}][OH^-]^2$$

By titrating a **saturated solution** of $Ca(OH)_2$ with standardised HCl, you can determine $[OH^-]$, then calculate $[Ca^{2+}]$, and hence K_{sp}.

Method

1. Prepare a **saturated solution** of $Ca(OH)_2$: add excess solid $Ca(OH)_2$ to distilled water in a stoppered flask. Shake and leave for at least 24 hours to reach equilibrium at a constant temperature.
2. **Filter** the saturated solution through filter paper to remove undissolved solid. The filtrate is a saturated $Ca(OH)_2$ solution (limewater).
3. Pipette **25.0 cm^3** of the filtrate into a conical flask. Add 2–3 drops of **methyl orange** indicator.
4. Titrate with standardised HCl from a burette until the indicator changes from **yellow to orange/red**.
5. Repeat for concordant results. Calculate the mean titre.

Calculation

1. Calculate moles of HCl used: n(HCl) = c(HCl) x V(titre in dm^3).
2. Using the equation $Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$: $n(Ca(OH)_2) = n(HCl) / 2$.
3. Calculate $[Ca(OH)_2] = n(Ca(OH)_2) / 0.0250$.
4. Since $Ca(OH)_2 \rightarrow Ca^{2+} + 2OH^-$: $[Ca^{2+}] = [Ca(OH)_2]$ and $[OH^-] = 2[Ca(OH)_2]$.
5. **$K_{sp} = [Ca^{2+}] \times [OH^-]^2$**. Substitute values. Literature value: ~5.5×10^{-6} mol^3 dm^{-9} at 25 degrees C.

Error Analysis

Source of Error	Type	Improvement
Solution not truly saturated (equilibrium not reached)	Systematic ($[OH^-]$ too low; K_{sp} too low)	Use excess solid; leave for 24+ hours; control temperature
Undissolved $Ca(OH)_2$ particles passing through filter	Systematic ($[Ca^{2+}]$ overstated as particles dissolve during titration)	Use fine grade filter paper; filter twice if necessary

CO_2 from air absorbed into the $Ca(OH)_2$ solution	Systematic (forms insoluble $CaCO_3$; removes Ca^{2+} and OH^-)	Keep the solution stoppered; prepare fresh; filter immediately before use
Temperature not controlled (K_{sp} varies with T)	Systematic	Record temperature; work at 25 degrees C using a thermostatted water bath

Exam Tip: Remember the stoichiometry: 1 mol $Ca(OH)_2$ reacts with **2 mol HCl**. Since $Ca(OH)_2$ dissociates into 1 Ca^{2+} and **2 OH^-**, the $[OH^-]$ is **twice** $[Ca(OH)_2]$. Do not forget to **square** $[OH^-]$ in the K_{sp} expression. K_{sp} only applies to **saturated** solutions at equilibrium. Units: $mol^3\ dm^{-9}$ (not the same as simple K_c units).

CP10: Electrochemical Cells and EMF

See the AQA RP10 entry for comprehensive theory, method, and error analysis of constructing electrochemical cells. Key formula: $\mathbf{E^o_{cell} = E^o_{cathode} - E^o_{anode}}$. The more reactive metal is the anode (oxidation). Use a high-resistance voltmeter to minimise current flow.

Exam Tip: Electrons flow from the more reactive metal (anode, more negative E^o) to the less reactive metal (cathode, more positive E^o) through the external wire. The salt bridge completes the internal circuit. A positive E^o_{cell} indicates a thermodynamically feasible reaction.

CP11: Temperature and Rate (Sodium Thiosulfate + HCl)

See the AQA RP3 entry for the detailed disappearing cross method. Rate = 1/t. Plot 1/t vs temperature for the rate-temperature relationship. Plot ln(1/t) vs 1/T for an Arrhenius plot (gradient = $-E_a/R$).

Exam Tip: For the Arrhenius plot: **$\ln k = -E_a/(RT) + \ln A$**. A straight-line graph of ln(rate) vs 1/T has gradient = $-E_a/R$. Convert temperature to **Kelvin** (K = degrees C + 273). Calculate E_a = −gradient x 8.314. Units of E_a: J mol^{-1} (divide by 1000 for kJ mol^{-1}).

CP12: Factors Affecting K_c (Le Chatelier's Principle)

See the AQA RP12 entry for the cobalt chloride equilibrium demonstration. Key Edexcel points: understand that **temperature is the only factor that changes the value of K_c**. Concentration changes shift the position of equilibrium but K_c remains constant. A catalyst does NOT change K_c or the equilibrium position.

Exam Tip: For an endothermic forward reaction: increasing temperature shifts equilibrium right and **increases K_c**. For an exothermic forward reaction: increasing temperature shifts equilibrium left and **decreases K_c**. Adding more of a reactant shifts equilibrium right but K_c stays the same (the system adjusts concentrations to restore K_c).

CP13: Prepare a Pure Organic Solid and Test Its Purity

Background Theory

Organic solids (e.g. aspirin, a substituted amide) are prepared by organic synthesis reactions, then purified by **recrystallisation** and their purity assessed by **melting point determination**. A pure substance has a **sharp melting point** (melts over a narrow range, typically <2 degrees C). An

impure substance melts over a wider range and at a lower temperature than expected (melting point depression).

Recrystallisation Method

1. **Dissolve** the crude product in the **minimum volume** of hot solvent (just enough to dissolve it at the boiling point). Common solvents: water, ethanol, ethyl ethanoate. The choice of solvent is critical: the product must be soluble in the hot solvent but insoluble in the cold solvent.
2. **Hot filter** (if necessary): filter the hot solution through a pre-warmed funnel with fluted filter paper. This removes insoluble impurities. Work quickly to prevent premature crystallisation.
3. **Cool slowly:** Allow the filtrate to cool slowly to room temperature, then place in an ice bath. Slow cooling produces larger, purer crystals (rapid cooling gives small, impure crystals because impurities get trapped).
4. **Filter under reduced pressure** (vacuum filtration using a Buchner funnel and flask). This is faster than gravity filtration and removes more solvent. The crystals are collected on the filter paper; the impurities remain dissolved in the cold filtrate (mother liquor).
5. **Wash** the crystals on the filter with a small volume of ice-cold solvent (to remove residual impurities from the crystal surfaces without re-dissolving the product).
6. **Dry** the crystals: press between sheets of filter paper, then leave in a desiccator (containing a drying agent like silica gel) or a warm oven (below the melting point).

Melting Point Determination

1. Pack a small amount of dry crystals into a **melting point capillary tube** (sealed at one end). The column of solid should be ~5 mm high.
2. Place the tube in an electrically heated **melting point apparatus** or attach it to a thermometer in a heated oil bath.
3. Heat slowly (~1–2 degrees C per minute near the expected melting point). Record the temperature at which the solid **first starts to melt** (meniscus appears) and the temperature at which it is **completely liquid**.
4. A **pure substance** melts over a narrow range (≤2 degrees C) and matches the literature value. An **impure substance** melts at a lower temperature and over a wider range.

Safety

- Hot solvents — risk of burns. Use a water bath to heat flammable solvents (never a naked flame).
- Organic solvents (ethanol, ethyl ethanoate) are flammable and produce vapours — good ventilation.
- Some organic products are irritants — wear gloves and goggles.

Calculating Percentage Yield and Atom Economy

Percentage yield = (actual moles of product / theoretical moles of product) x 100
Reasons yield < 100%: incomplete reaction, side reactions, loss during transfer/filtration/washing, product left in mother liquor.

Atom economy = (M_r of desired product / sum of M_r of all products) x 100
A measure of how efficiently atoms are used. Higher atom economy = less waste.

Error Analysis

Source of Error	Type	Improvement
Product lost during recrystallisation (stays in solution)	Systematic (lowers yield)	Use the minimum volume of solvent; cool fully to maximise crystal recovery
Impurities not fully removed	Systematic (melting point too low and broad)	Repeat the recrystallisation; use a better solvent choice
Product not fully dry	Systematic (mass too high; melting point too low)	Dry thoroughly in desiccator; check mass constancy by re-weighing
Heating too fast during melting point	Systematic (overshoot; temperature reads too high)	Heat at 1–2 degrees C per minute near the expected melting point

Exam Tip: In recrystallisation: **minimum volume** of hot solvent ensures maximum yield. **Slow cooling** gives larger, purer crystals. **Vacuum filtration** (Buchner) is faster and gives drier

crystals than gravity filtration. A melting point range of >2 degrees C indicates impurity. Always compare your melting point with the literature value.

CP14: Prepare a Pure Organic Liquid (Reflux and Distillation)

Background Theory

Many organic reactions require **prolonged heating** to go to completion, but the reagents and products are often volatile and would be lost if heated in an open flask. **Reflux** solves this: a vertical condenser is attached to the flask so that vapours condense and drip back into the reaction mixture, allowing continuous heating without loss of material. After the reaction is complete, the product is separated by **distillation**.

See the AQA RP5 entry for the detailed distillation method. Key additional points for organic liquid preparation:

Reflux Method

1. Place reagents in a round-bottomed flask. Add anti-bumping granules.
2. Attach a **vertical condenser** (open at the top — not sealed) to the flask. Water flows in at the bottom and out at the top.
3. Heat the mixture gently (water bath for flammable liquids, or electric heating mantle). Vapours rise, condense in the condenser, and **return to the flask**. Continue for the required time (e.g. 30–60 minutes).
4. After reflux, rearrange the apparatus for **distillation**: tilt the condenser to an angle, connect to a collecting flask, and distil the product at its boiling point.

Purification Steps (After Distillation)

1. **Washing:** Use a separating funnel to wash the organic product with aqueous solutions to remove impurities. For example, wash with $NaHCO_3$ (removes acidic impurities) or dilute acid (removes basic impurities). Shake, vent, and drain off the aqueous layer.

2. **Drying:** Add an anhydrous drying agent (e.g. anhydrous $MgSO_4$, $CaCl_2$, or Na_2SO_4) to the organic layer. Leave for 10–15 minutes, swirling occasionally. The drying agent absorbs residual water. Filter off the drying agent by gravity filtration.
3. **Redistillation:** Distil the dried product, collecting the fraction at the correct boiling point (±2 degrees C). This gives the purest product.

Purity Testing

- **Boiling point:** A pure liquid has a sharp, constant boiling point matching the literature value. Impurities raise the boiling point and cause it to boil over a range.
- **Refractive index:** Measured using a refractometer. Each pure liquid has a characteristic refractive index. Impurities alter the value.
- **GC (gas chromatography):** A pure substance shows a single peak. Impurities appear as additional peaks.

Safety

- Organic liquids are **flammable** — use electric heating (water bath or heating mantle). No naked flames.
- Never heat a sealed system — the condenser must be open at the top during reflux.
- When using a separating funnel: **vent regularly** by inverting and opening the tap to release gas pressure build-up.
- Some drying agents (anhydrous $CaCl_2$) react with alcohols and cannot be used for drying them.

Exam Tip: Reflux = heating without losing vapour (condenser vertical, vapour returns to flask). Distillation = separating by boiling point (condenser angled, vapour collected). The thermometer bulb should be at the **side-arm outlet** during distillation. When using a separating funnel, the organic layer may be on top (less dense than water) or bottom (if denser, e.g. DCM). Know which layer to keep! For drying: anhydrous Na_2SO_4 is a general-purpose drying agent suitable for most organic liquids.

© RAG Learning · Not for redistribution

CP15: Tests to Identify Functional Groups

See the AQA RP6 entry for comprehensive functional group tests: bromine water for alkenes, Tollens' and Fehling's for aldehydes, Na_2CO_3 for carboxylic acids, $K_2Cr_2O_7/H^+$ for primary/secondary alcohols. Key distinction: Tollens'/Fehling's test for aldehydes but NOT ketones (aldehydes can be oxidised further; ketones cannot).

Additional Edexcel Tests

Functional Group	Test	Positive Result
Halogenoalkane (C–X)	Add $AgNO_3$ after hydrolysis with NaOH(aq) and warm. Acidify with HNO_3	White ppt (Cl), cream ppt (Br), yellow ppt (I)
Phenol (–OH on benzene ring)	Add neutral $FeCl_3$(aq)	Purple/violet colour (iron(III) phenoxide complex)
Ester (–COO–)	Warm with NaOH(aq) (hydrolysis), then acidify and smell	Sweet/fruity smell disappears on hydrolysis; no effervescence with Na_2CO_3
Primary amine (–NH_2)	React with nitrous acid ($NaNO_2$ + HCl) at <10 degrees C, then add 2-naphthol	Orange/red azo dye precipitate (diazo coupling)

Exam Tip: For the halogenoalkane test, the rate of hydrolysis (and hence precipitate formation) depends on the C–X bond strength: C–I reacts fastest (weakest bond), C–Cl slowest (strongest bond). This is often tested as a comparison experiment: add $AgNO_3$ to hydrolysed solutions and time the appearance of precipitates.

© RAG Learning · Not for redistribution

CP16: pH Changes During Acid-Base Titrations

See the AQA RP9 entry for the full method, titration curve features, and analysis. Key Edexcel points:

Four Titration Curve Types

Titration	Initial pH	Equivalence pH	Indicator	Curve Shape
Strong acid + strong base	~1	7	Methyl orange or phenolphthalein	Steep vertical section at equivalence
Strong acid + weak base	~1	<7	Methyl orange only	Steep but shorter vertical section
Weak acid + strong base	~3	>7	Phenolphthalein only	Buffer region; half-equivalence: pH = pK_a
Weak acid + weak base	~3	~7	No suitable indicator (use pH meter)	Very gradual change; no distinct vertical section

Key Concepts

- **Buffer region:** The flat section before equivalence in a weak acid/strong base titration. The mixture of HA and A^- resists pH changes.
- **Half-equivalence point:** $[HA] = [A^-]$, so pH = pK_a. Read directly from the graph.
- **Equivalence point:** All the acid has reacted. For weak acid/strong base, pH > 7 (conjugate base hydrolysis). For strong acid/weak base, pH < 7.
- **Indicator choice:** The indicator's colour-change range must fall within the steep vertical section of the curve. If it does not, the end-point will not coincide with the equivalence point.

Exam Tip: You must be able to **sketch all four titration curves** and label key features (initial pH, buffer region, half-equivalence, equivalence point, final pH). At the half-equivalence point

for a weak acid: **$pH = pK_a$**. This is the easiest experimental method to determine pK_a.

Henderson-Hasselbalch equation: $pH = pK_a + \log([A^-]/[HA])$.